

Systematic Exploration of OAK’s Normalising Performance on Synthetic Datasets

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1 Introduction and Purpose

This document summarizes a systematic grid-based exploration to investigate how the Orthogonal Additive Kernel (OAK) Gaussian Process model assigns Sobol indices across different synthetic datasets.

The datasets vary across key dimensions:

- **Distribution:** Normal, Uniform, and Beta distributions.
- **Mean Location:** Centered (mean = 0) or shifted (mean = 6, 12, etc.).
- **Variance:** Low (variance = 1) or high (variance = 12).
- **Symmetry:** Symmetric distributions (centered around 0) versus asymmetric distributions (shifted to positive or negative ranges).

The goal is to understand how these dataset properties influence OAK’s ability to correctly allocate variance across Sobol indices after having normalised them using its inbuilt normalising flow, especially when the true data generating process is dominated by 4th order interactions of the form (as these have not been seen in the Revisited paper, which only has orders up to 3rd):

$$y = x_1 \cdot x_2 \cdot x_3 \cdot x_4$$

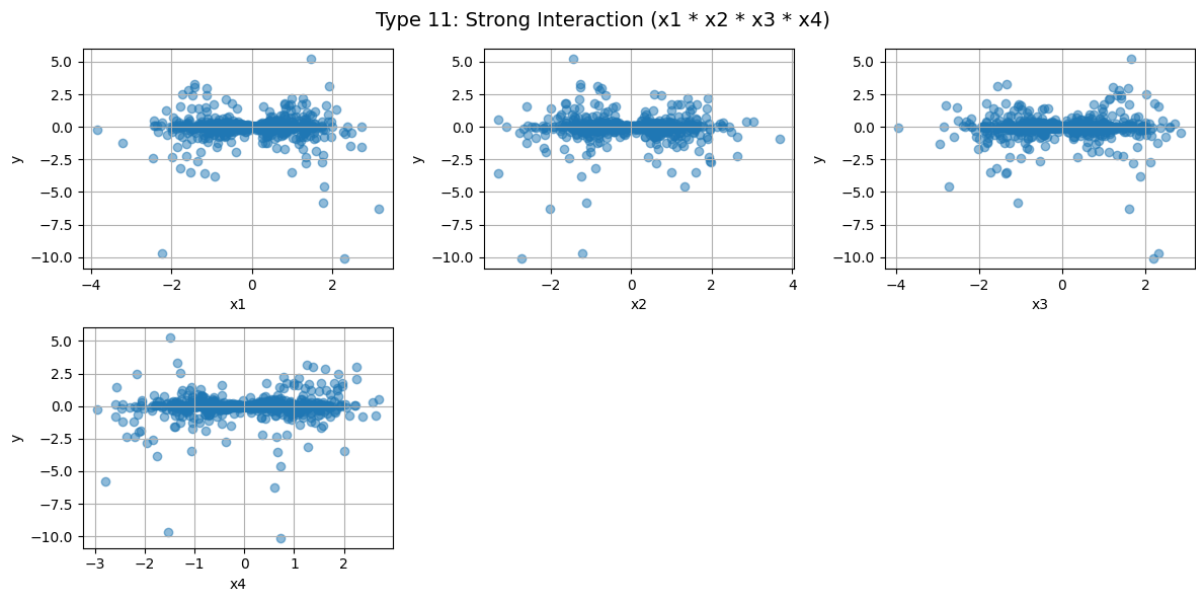
Each section below corresponds to one dataset configuration.

2 Results by Dataset Configuration

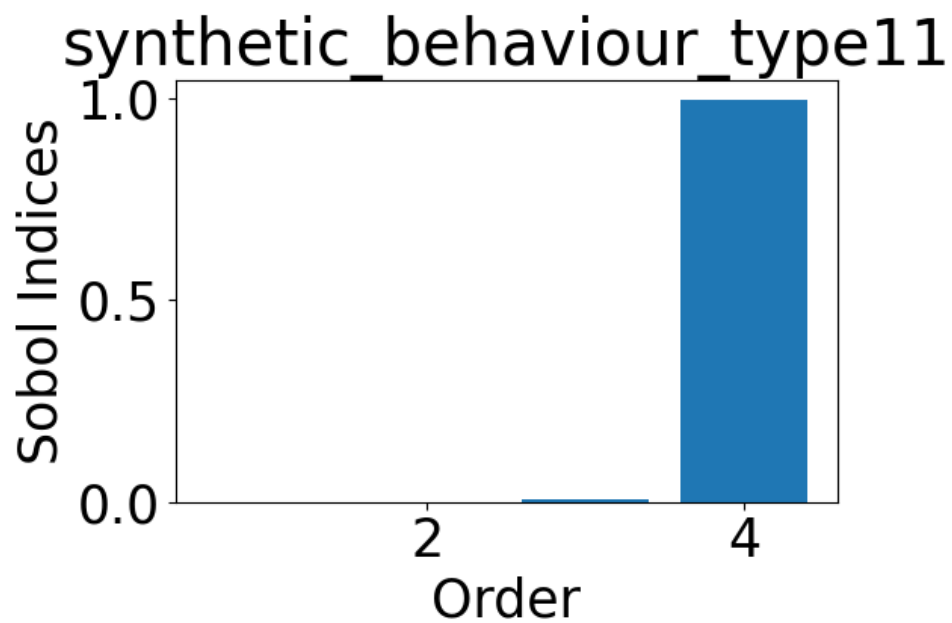
2.1 Centered Normal, Low Variance (Variance = 1)

X Generation:

$x_1, x_2, x_3, x_4 \sim N(0, 1)$
 $y = x_1 * x_2 * x_3 * x_4$



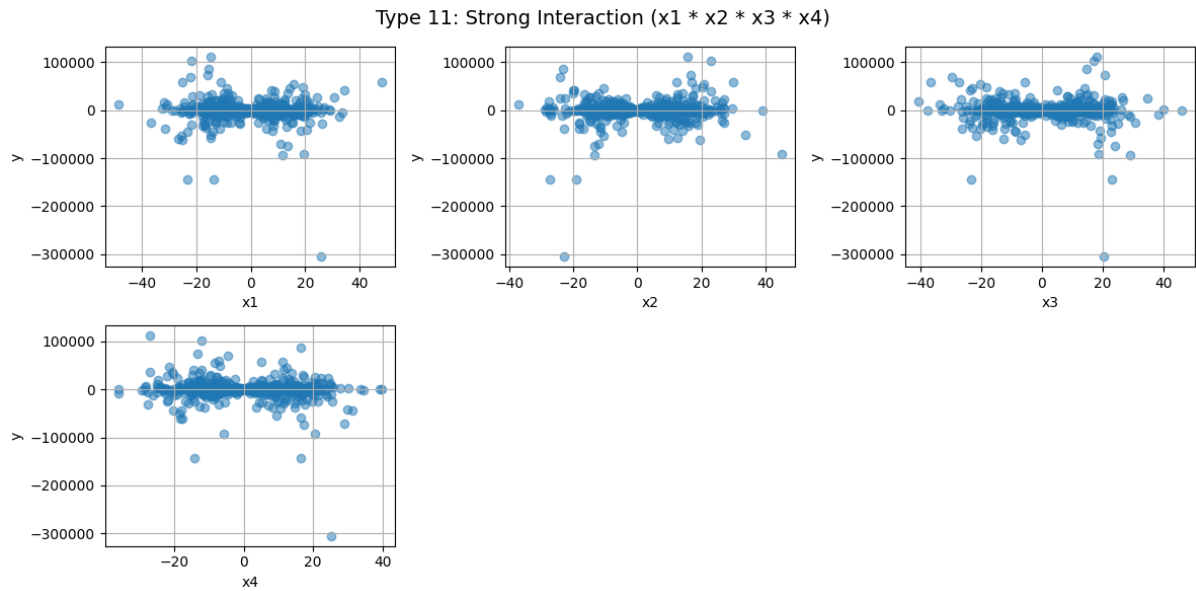
Sobol Indices Plot:



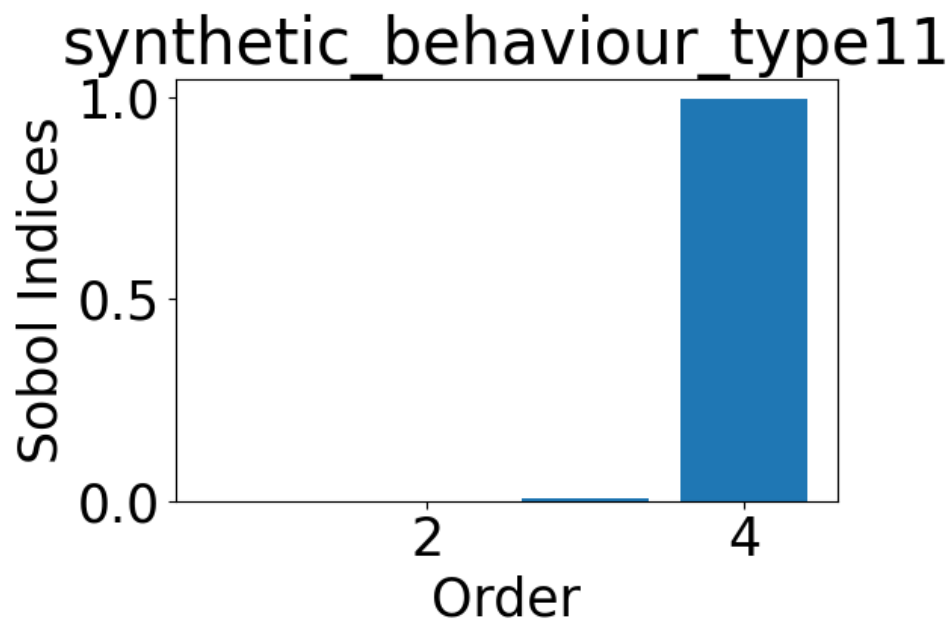
2.2 Centered Normal, High Variance (Variance = 12)

X Generation:

$$x_1, x_2, x_3, x_4 \sim N(0, 12)$$



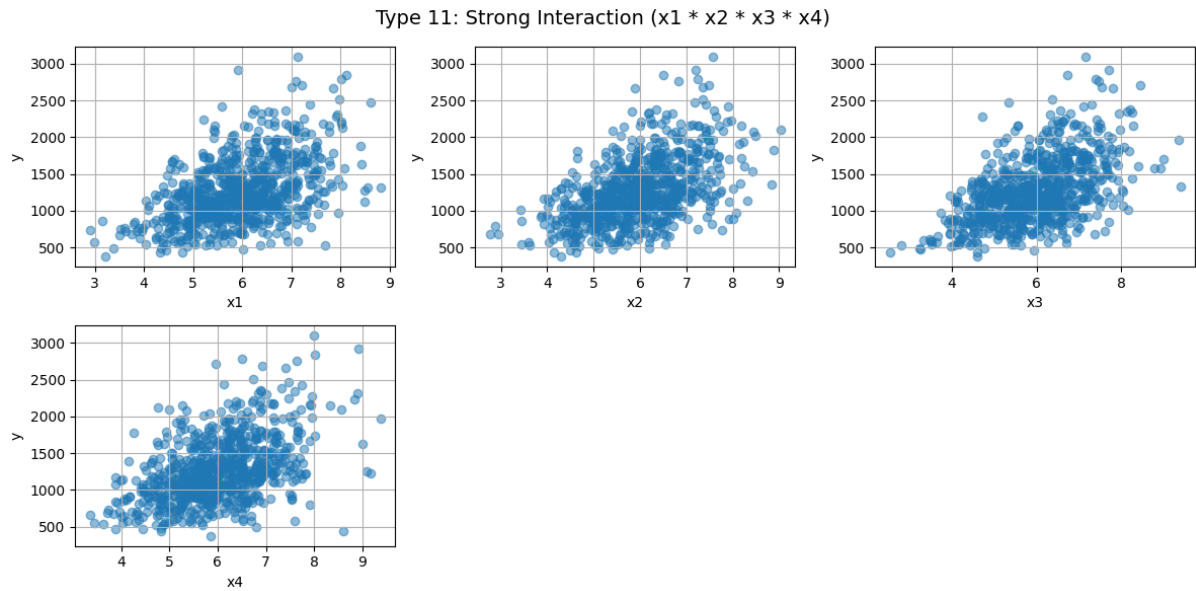
Sobol Indices Plot:



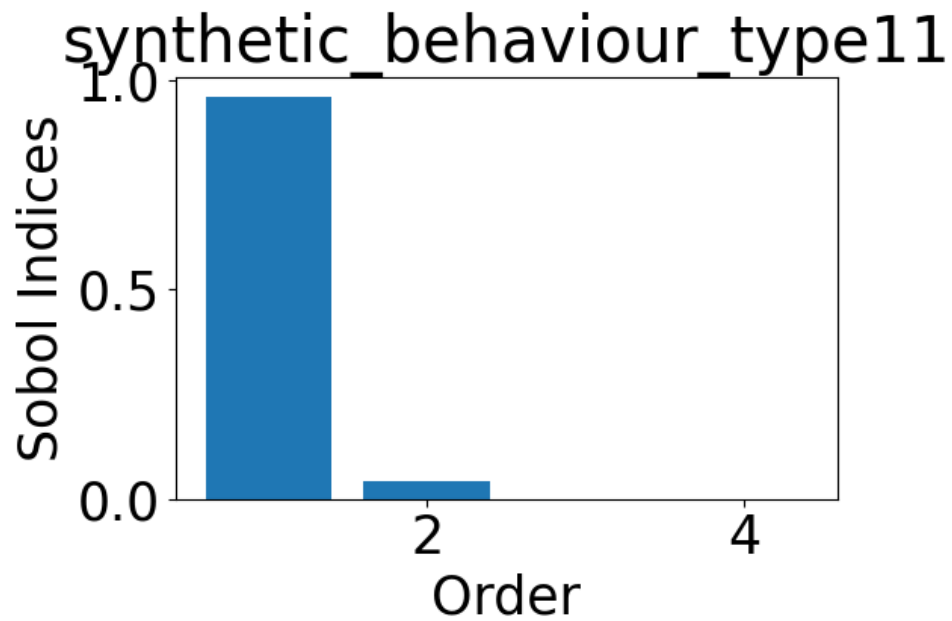
2.3 Shifted Normal, Low Variance (Mean = 6, Variance = 1)

X Generation:

$$x_1, x_2, x_3, x_4 \sim N(6, 1)$$



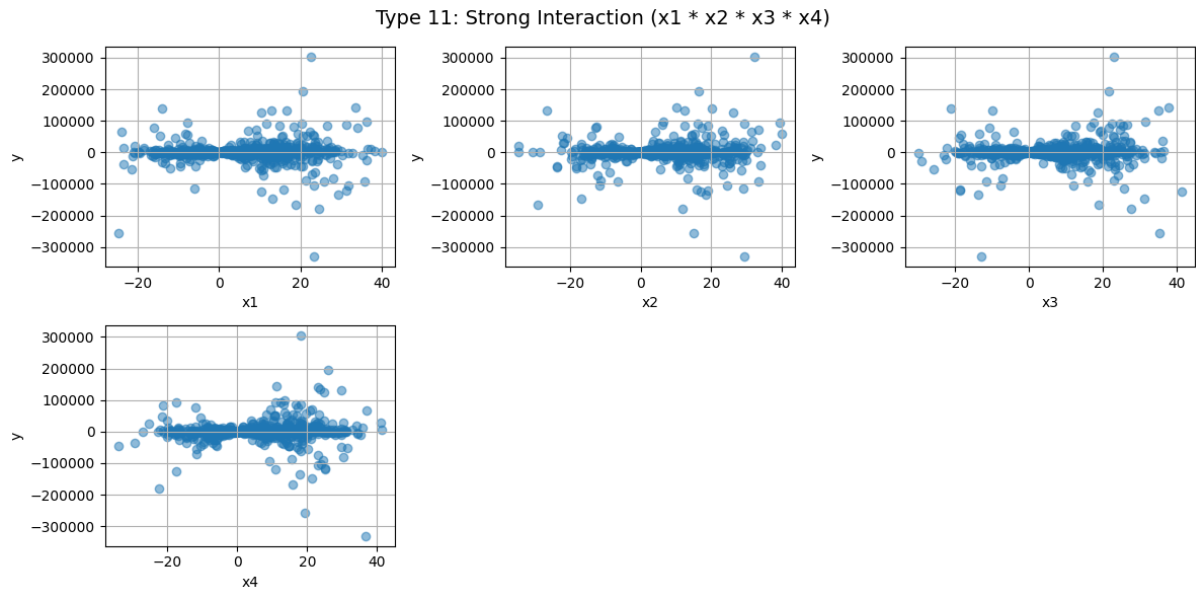
Sobol Indices Plot:



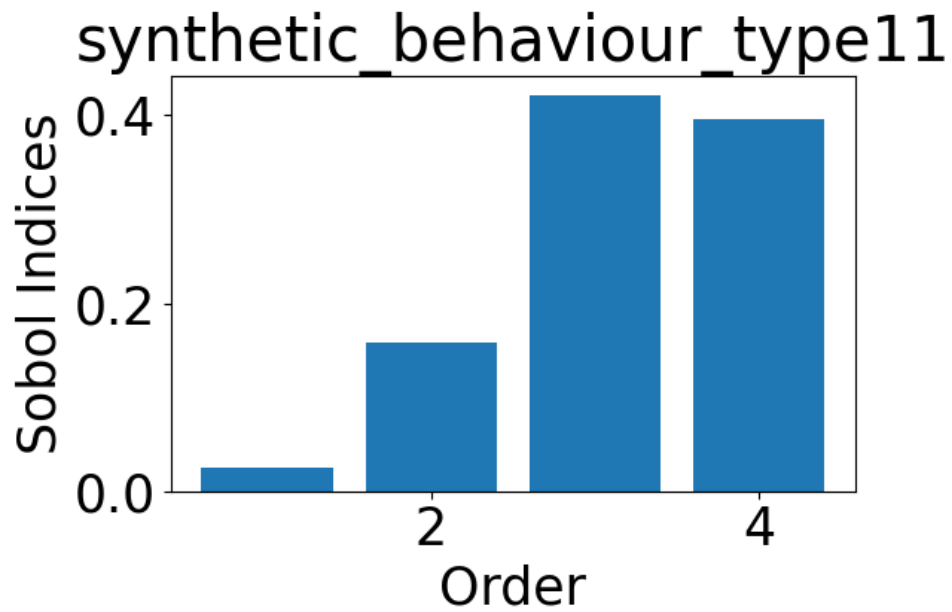
2.4 Shifted Normal, High Variance (Mean = 6, Variance = 12)

X Generation:

$$x_1, x_2, x_3, x_4 \sim N(6, 12)$$



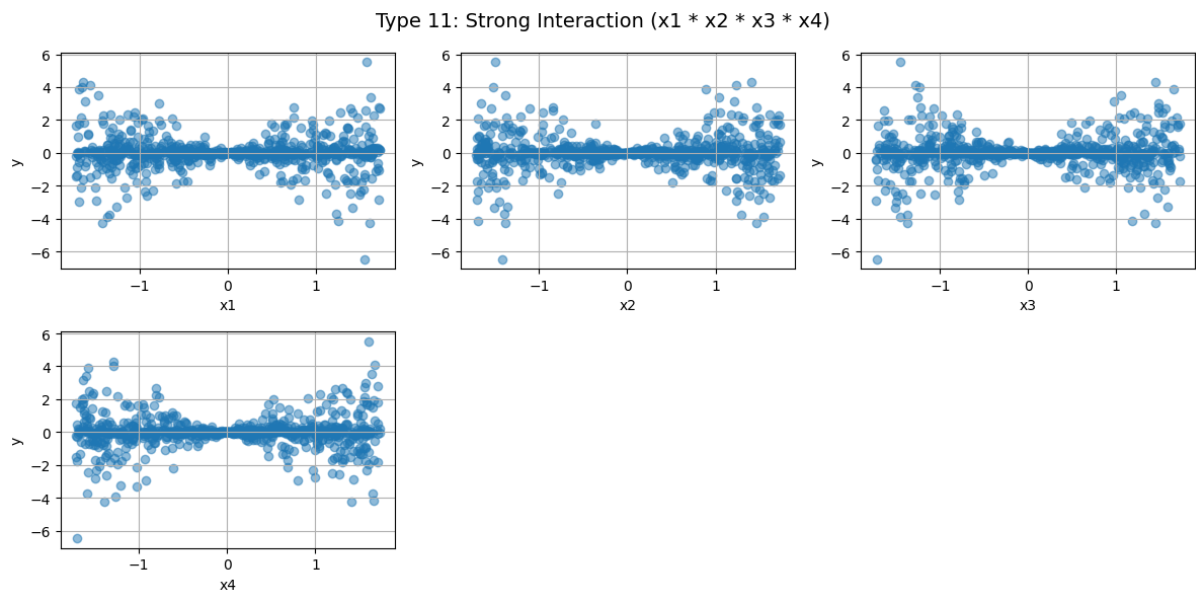
Sobol Indices Plot:



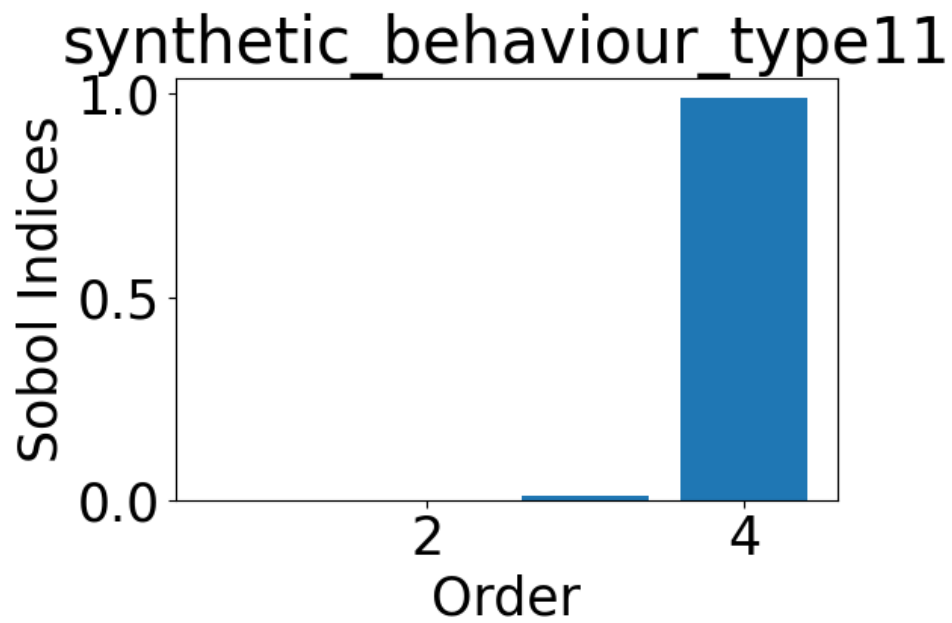
2.5 Centered Uniform, Low Variance (Variance = 1)

X Generation:

$x_1, x_2, x_3, x_4 \sim U(-\sqrt{3}, \sqrt{3})$



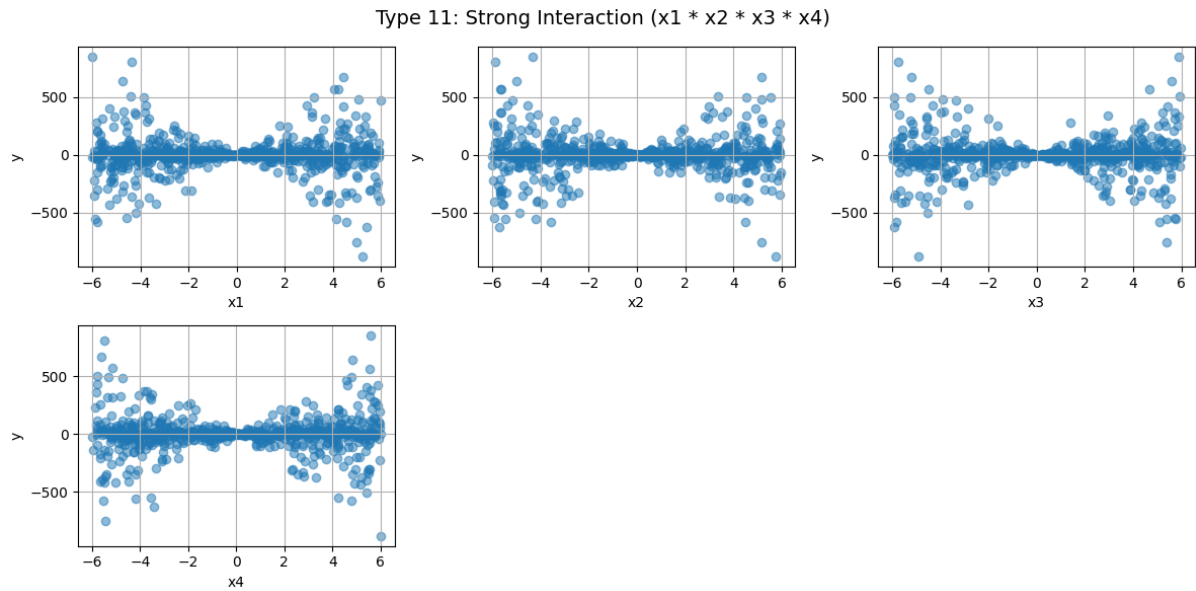
Sobol Indices Plot:



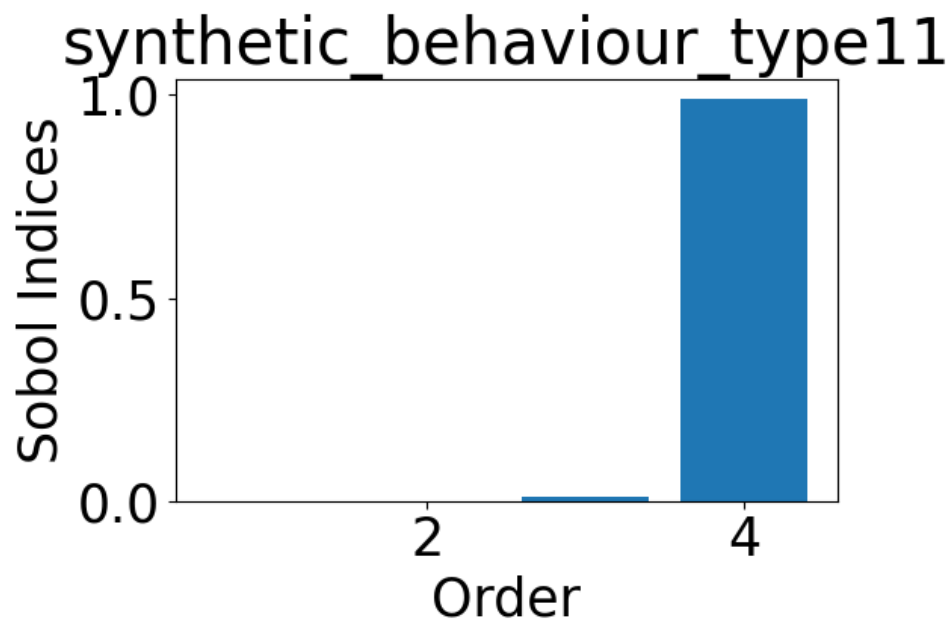
2.6 Centered Uniform, High Variance (Variance = 12)

X Generation:

$x_1, x_2, x_3, x_4 \sim U(-6, 6)$



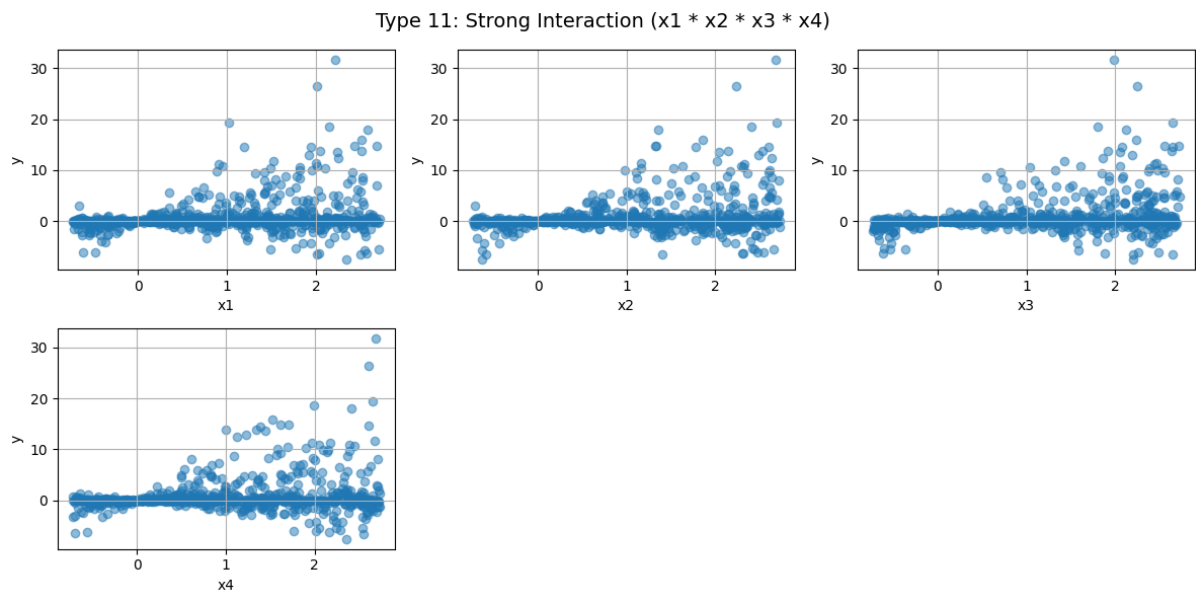
Sobol Indices Plot:



2.7 Shifted Uniform, Low Variance (Mean = 1, Variance = 1)

X Generation:

$x_1, x_2, x_3, x_4 \sim U(1-\sqrt{3}, 1+\sqrt{3})$



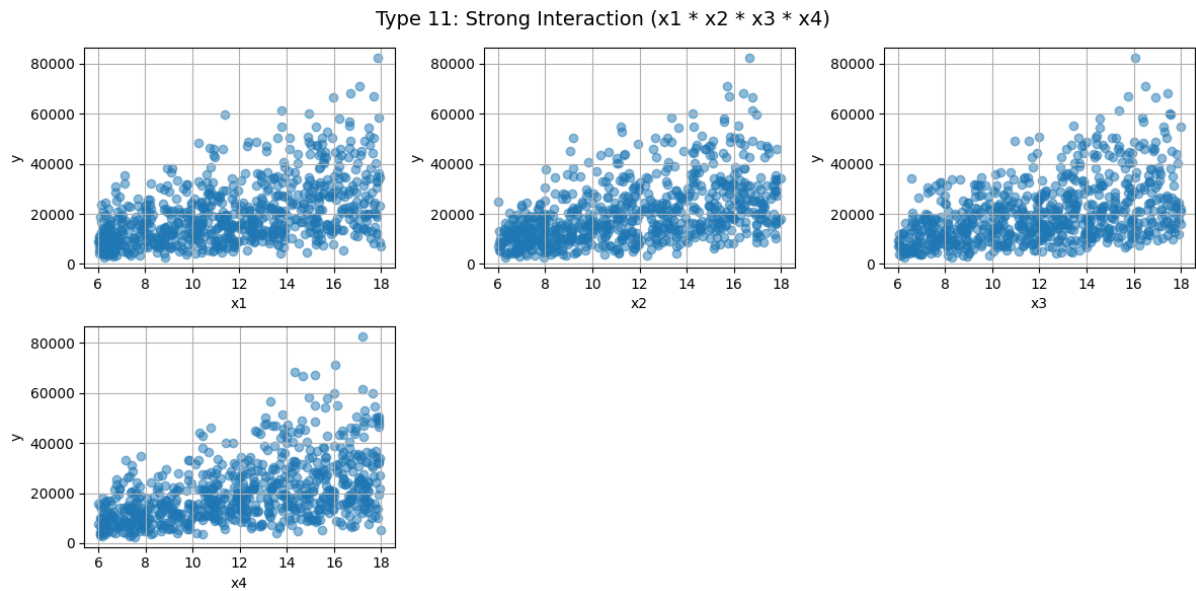
Sobol Indices Plot:



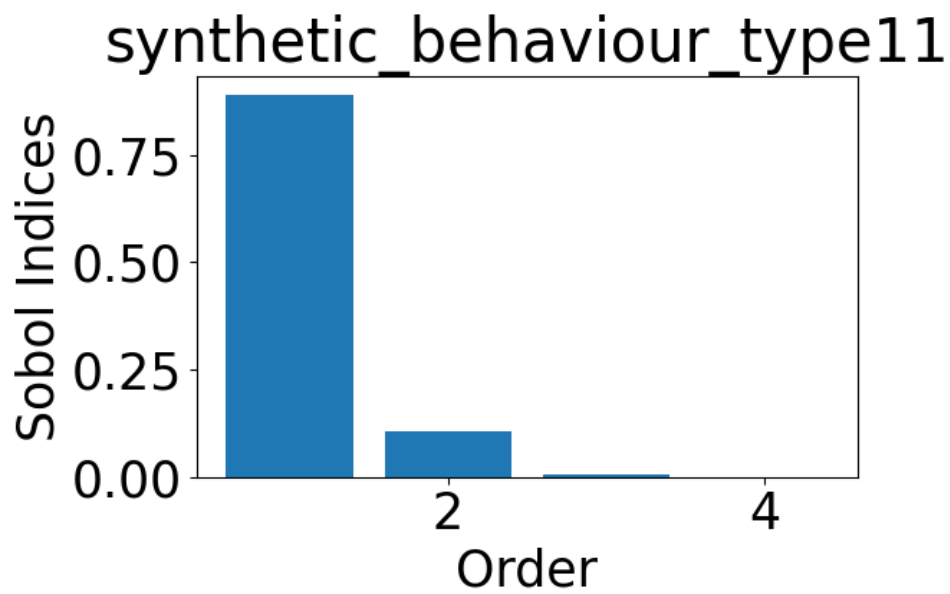
2.8 Shifted Uniform, High Variance (Mean = 12, Variance = 12)

X Generation:

$x_1, x_2, x_3, x_4 \sim U(6, 18)$



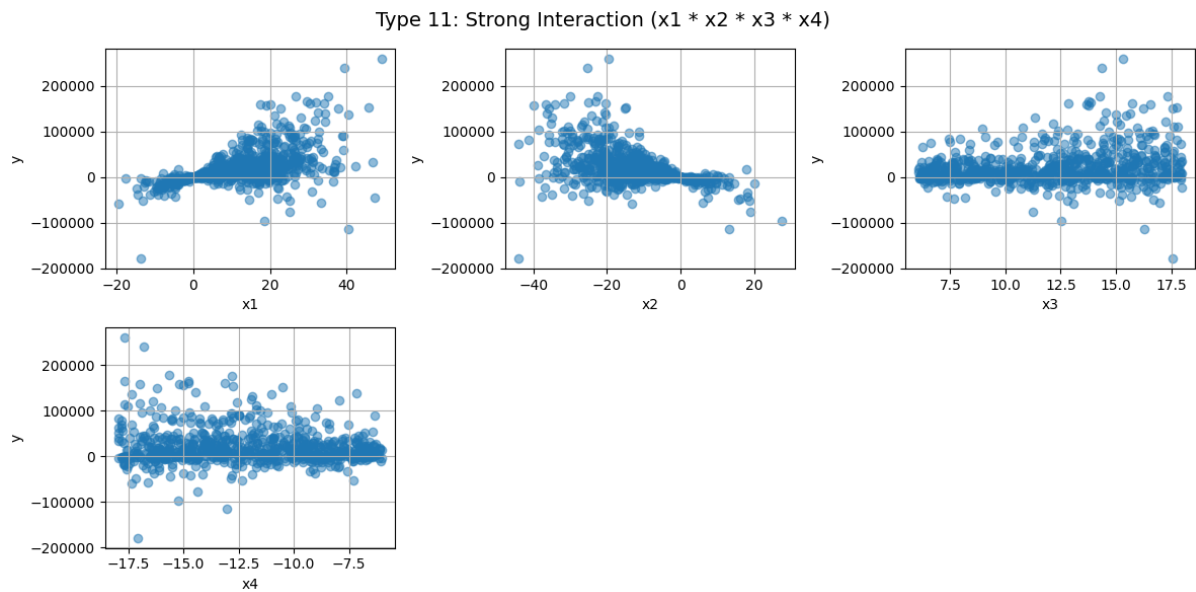
Sobol Indices Plot:



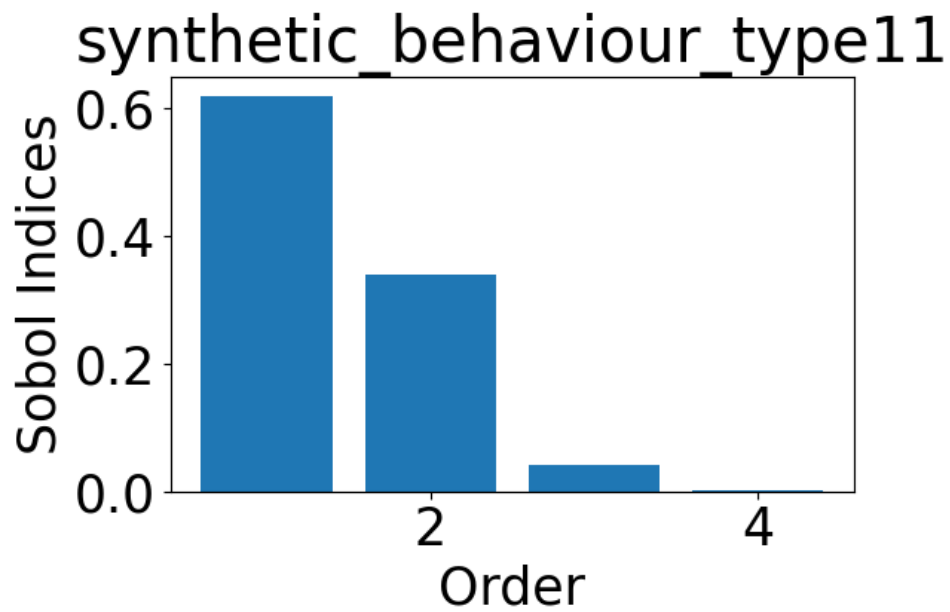
2.9 Mixed Distribution (Normal + Uniform)

X Generation:

$x_1 \sim N(12, 12)$
 $x_2 \sim N(-12, 12)$
 $x_3 \sim U(6, 18)$
 $x_4 \sim U(-18, -6)$



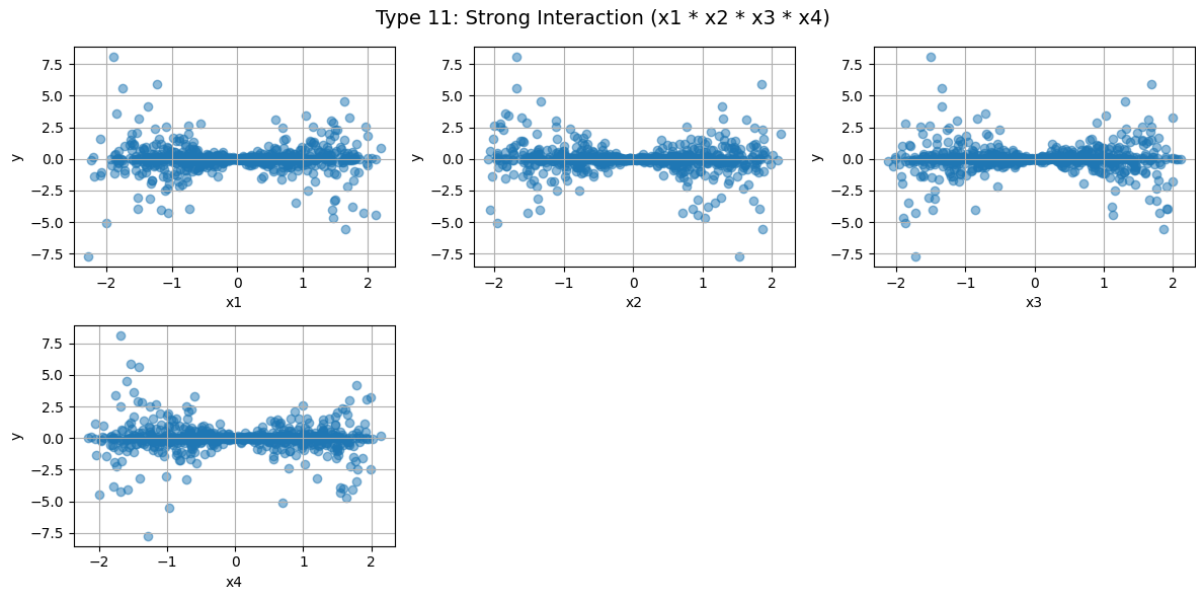
Sobol Indices Plot:



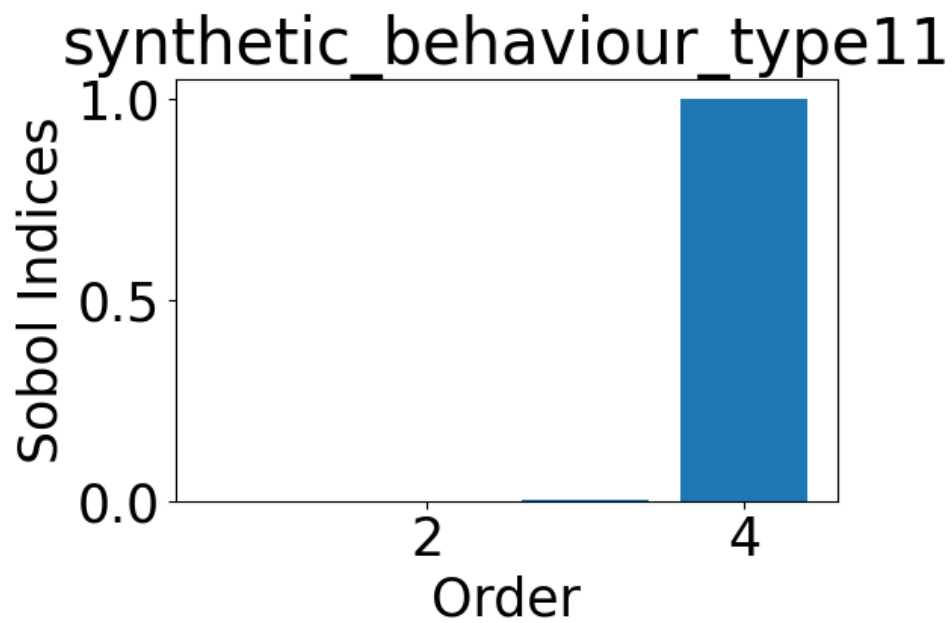
2.10 Symmetric Beta (Variance = 1)

X Generation:

$x_1, x_2, x_3, x_4 \sim \text{Beta}(2, 2)$, rescaled to mean 0, variance 1



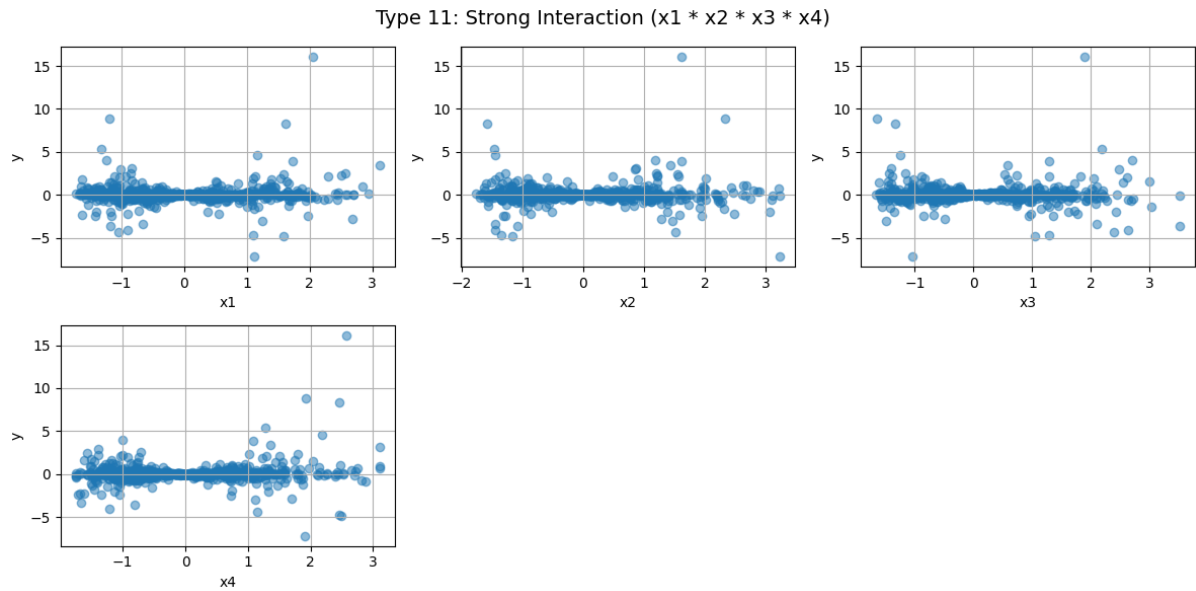
Sobol Indices Plot:



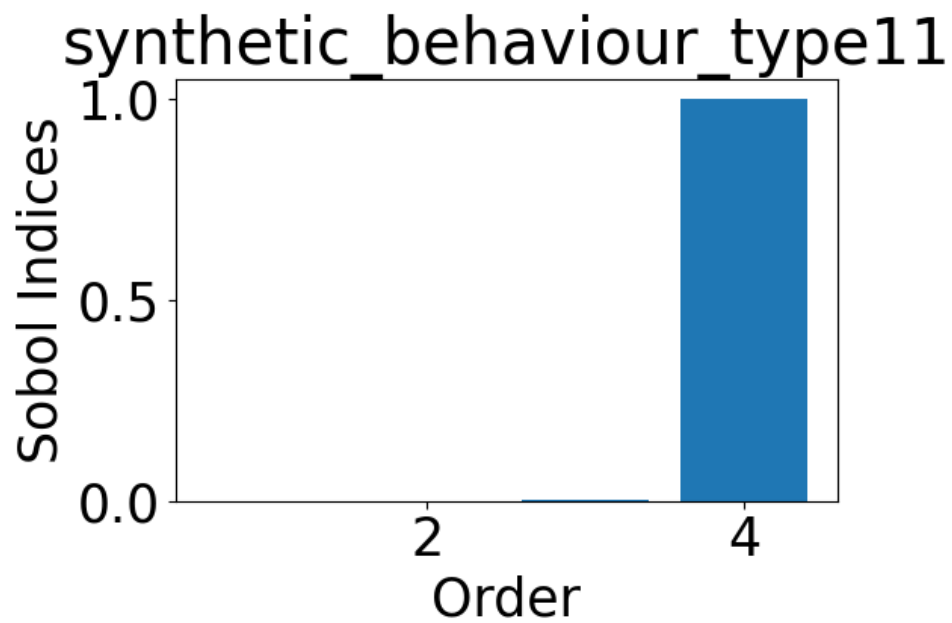
2.11 Asymmetric Beta (Variance = 1)

X Generation:

$x_1, x_2, x_3, x_4 \sim \text{Beta}(2, 5)$, rescaled to mean 0, variance 1



Sobol Indices Plot:



3 Feature Normalisation in OAK Model Fitting

The `fit()` function of the OAK model performs normalisation to ensure the model operates effectively across heterogeneous input features. The approach varies depending on the type of each feature:

- **Continuous Features:** These features are either processed using *normalising flows* (if `use_normalising_flow=True`) or standardised using a *Z-score transformation* (mean zero, unit variance) if `use_normalising_flow=False`. This boolean is set when the model is initialised. The normalising flow is a flexible bijective transformation that maps the data into a standard Gaussian space by minimising a Kullback-Leibler (KL) divergence objective.
- **Binary Features:** No normalisation is applied to binary features. These features are retained in their original form, ensuring their binary nature (0 or 1) is preserved.
- **Categorical Features:** Similarly, categorical features are left untransformed, preserving their integer-encoded categories throughout the training process.
- **Features with Empirical Measures:** For specific continuous features identified for *empirical measures*, standardisation using *Z-score transformation* is applied. This avoids numerical instability when such features are used with empirical distributions in the kernel.
- **Features Modelled with Gaussian Mixture Models (GMM):** Continuous features for which Gaussian Mixture Models (GMM) are fitted are excluded from further normalisation. The GMM parameters themselves capture the necessary statistical properties of these dimensions.
- **Target Variable (Y):** The target variable is always normalised using a *Z-score transformation* to ensure numerical stability in the Gaussian process likelihood.

The transformations for continuous features (whether normalising flows or standardisation) are learned directly from the training data during fitting. All transformations are applied before the model is constructed and trained. Importantly, binary and categorical features are preserved exactly, ensuring no unintended distortions are introduced for discrete variables.